Cartesian Equation of Lines in R^2

Knowledge/ Understanding

3. For each of the following, find the scalar equation of the line that passes through the point P_0 and has normal vector $\vec{n}$.
   a. $P_0(4, -2), \vec{n} = (2, 7)$
   b. $P_0\left(\frac{1}{2}, 2\right), \vec{n} = (-4, 0)$
   c. $P_0(3, 3), \vec{n} = (1, 1)$
   d. $P_0\left(\frac{1}{3}, \frac{1}{3}\right), \vec{n} = (-1, 1)$

4. For each of the following, find a normal vector, a direction vector, and a point on each line.
   a. $4x + 3y - 12 = 0$
   b. $3x - 6y = 14$
   c. $x = 5$
   d. $y = 3x - 10$

5. Prove that both $(-b, a)$ and $(b, -a)$ are perpendicular to (a, b) for all a and b.

6. Find the Cartesian equation of each of the following lines.
   a. $(x, y) = (4, -6) + t(8, 2)$
   b. $x = 3 + 18t, y = 4 + 9t$
   c. $\vec{r} = (2, 7) + t(2, 7)$
   d. $x = 2t, y = -2$

7. Find the scalar equation of the line that passes through $(2, -6)$ and
   a. is parallel to $2x - 3y + 8 = 0$
   b. is perpendicular to $3x - 2y + 12 = 0$
   c. has a direction vector $(2, -3)$
   d. has a normal vector $(3, -2)$

8. Find the scalar equation of the line through $(8, -2)$ that is parallel to the line $x = -4 - 5t, y = 11 + 3t$ by first finding the symmetric equation of this line, and then simplifying it.

Part B

9. Find vector, parametric, and symmetric equations of the following lines.
   a. $5x - 3y + 15 = 0$
   b. $-4x + 6y + 9 = 0$

Answers

3. a. $2x + 7y + 6 = 0$ **b.** $2x - 1 = 0$ **c.** $x + y - 6 = 0$ **d.** $x - y = 0$ **4. a.** $\vec{n} = (4, 3), \vec{d} = (3, -4), P(3, 0)$ **b.** $\vec{n} = (1, -2); \vec{d} = (2, 1), P\left(\frac{14}{3}, 0\right)$ **c.** $\vec{n} = (1, 0); \vec{d} = (0, 1), P(5, 6)$ **d.** $\vec{n} = (3, -1); \vec{d} = (1, 3), P(2, -4)$ **6. a.** $x - 4y - 28 = 0$ **b.** $x - 2y + 5 = 0$ **c.** $7x - 2y = 0$ **d.** $y + 2 = 0$ **7. a.** $2x - 3y - 22 = 0$ **b.** $2x + 3y + 14 = 0$ **c.** $3x + 2y + 6 = 0$ **d.** $3x - 2y - 18 = 0$ **8.** $3x + 5y - 14 = 0$ **9. a.** $\vec{r} = (0, 5) + t(3, 5); x = 3t, y = 5t + 5; \frac{x}{3} = \frac{y - 5}{5}$ **b.** $\vec{r} = \left(0, \frac{-3}{2}\right) + t(3, 2); x = 3t, y = 2t - \frac{3}{2}; \frac{x}{3} = \frac{y + \frac{3}{2}}{2}$ **11. a.** $\frac{7}{\sqrt{13}}$ **b.** $\frac{12}{\sqrt{53}}$ **c.** 0 **d.** 8